AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 22 /April/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 hours

Instructions:

- Total Marks: **60**
- This exam consists of 5 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [10 Marks]

For each question, choose the most appropriate answer. Copy the answer to the External Answer booklet. Each question carries 2 marks.

1. Which of the following is NOT a valid Java data type?
   a. double
   b. float
   c. long
   d. real
2. Which of the following methods is used to read input from the user?
   a. `print()`
   b. `input()`
   c. `Scanner()`
   d. `nextDouble()`
3. Which of the following Boolean expressions will give true?
   a. `(3!=2+1)`
   b. `!(3==2+1)`
   c. `(3!=2+1)||( 3==2+1)`
   d. `(3!=2+1)&&( 3==2+1)`
4. What is the value of `a` after executing the following statement:
   ```
   int a;
   ```
   a. 0
   b. 1
   c. false
   d. undefined
5. Which operator can be used with numbers in Java?
   a. +
   b. -
   c. /
   d. All answers

Question 2 [10 marks]

The following code includes some errors. Find the errors and rewrite the code after correction.

```
import java.util.Scanner;
public class MTA {
    public static void main(String [] args) {
        Scanner input = new Scanner(System.in);
        int x = input.nextInt;
        if x > 10
            System.out.print(x+" ");
            System.out.println("is Greater than 10")
        else
            System.out.println(x" is Less than or equal to 10");
    }
}
```

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Question 3 [15 Marks]

Trace the following pieces of code and write their exact output:

a.

```
int x= 10;
int y = 3;
boolean z = x==y;
System.out.println(z);
System.out.println(y*x%3);
System.out.println((x+y)%310);
```

b.

```
for (int i = 2; i <= 9; i++) {
    System.out.println(i);
    i++;
}
```

c.

```
int x = 11;
if (x / 2 > 5) {
    x /= 2;
    System.out.println(x);
    x++;
} else {
    x *= 2;
    System.out.println(x);
    x--;
}
System.out.println(x);
```

d.

```
String s1,s2;
s1 = "Barcode ";
s2 = " KD";
System.out.println(s1+20+13+14);
System.out.println(10+20+s2);
```

For questions 4 and 5, copy the below code into your answer sheet and continue the rest of the code according to the given specifications of each question

```
import java.util.Scanner;
public class MTA {
      public static void main(String[] args) {

    }
}
```

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Question 4 [18 Marks]:

The number of credit hours needed to finish the ITC track at AOU is 131. The student's academic level is determined as follows:

Academic level	Number of credit hours finished
Level 1	Credit hours < 33
Level 2	33 <= Credit hours < 65
Level 3	65 <= Credit hours < 98
Level 4	Credit hours >=98

You need to write a Java program that works as follows:

- Read from the user an integer that represents the number of credit hours finished.
- Find and print the student's academic level category based on the above table.
- If the user entered a value of credit hours that is negative or more the 131, the program should print an error message.

Here are different samples of the program run to help you understand how the program works.

1	Enter the number of credit hours you have finished: -1 Error! Incorrect number of credit hours!
2	Enter the number of credit hours you have finished: 200 Error! Incorrect number of credit hours!
3	Enter the number of credit hours you have finished: 20 You are in Level 1
4	Enter the number of credit hours you have finished: 50 You are in Level 2
5	Enter the number of credit hours you have finished: 70 You are in Level 3
6	Enter the number of credit hours you have finished: 120 You are in Level 4

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Question 5 [7 Marks]:

Using a for loop, write a Java code to print the following sequence of numbers in one line separated by spaces:

2 5 8 11 14 17 20

End of questions

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